PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

Project Detail	Value
Date	21 October 2026
Team ID	Individual Project
Project Name	StaffSync – Modern Human Resource Management System
Maximum Marks	4 Marks

1. Project Overview

StaffSync is a MERN-stack Human Resource Management System designed for three user roles: HR, Manager, and Employee. It supports employee management, attendance tracking, leave requests and approvals, announcements, training programs, performance records, and notifications. The React frontend communicates with a Node.js/Express REST API, which uses MongoDB for persistent storage. JWT-based authentication and role-based authorization are intended to protect restricted operations.

2. Data Flow Diagram — Level 0 (Context Diagram)

This context diagram shows the principal data exchanges between Employees, Managers, HR, and the StaffSync HRMS.

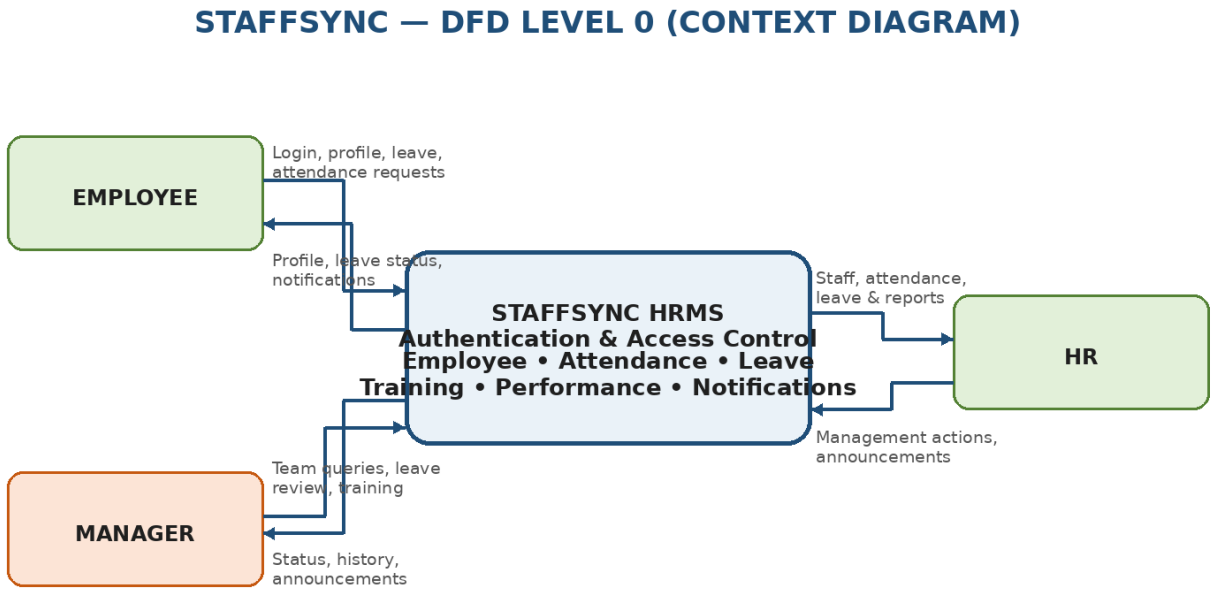


Figure 1. StaffSync Level 0 Data Flow Diagram (Context Diagram).

3. Data Flow Diagram — Level 1

This diagram expands StaffSync into major processes and identifies the MongoDB collections used as logical data stores.

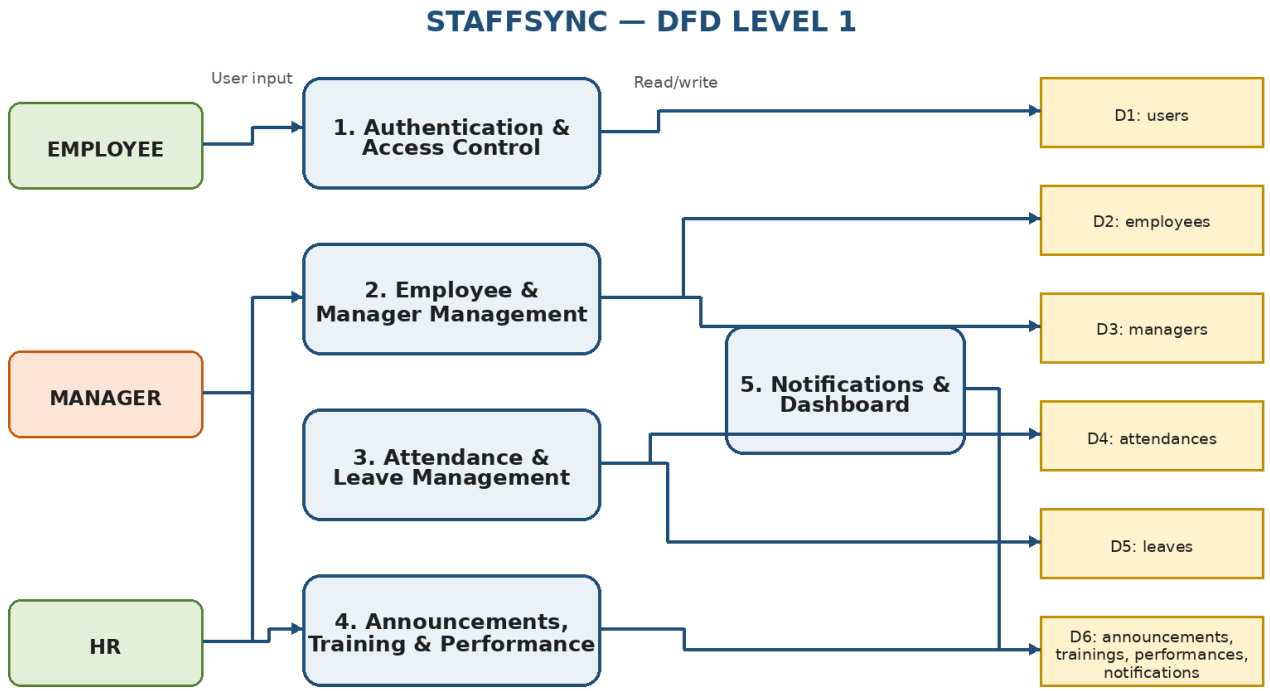


Figure 2. StaffSync Level 1 Data Flow Diagram.

4. Main Data Stores

Collection	Purpose
users	Authentication identity, email, password hash, and role.
employees	Employee profile and employment-related details.
managers	Manager-specific information and team relationships.
attendances	Daily attendance records and check-in/check-out information.
leaves	Leave applications, review decisions, and status.
announcements	HR announcements visible to permitted users.
trainings	Training programs and related participation information.
notifications	User-facing alerts about relevant system events.
performances	Performance reviews, ratings, goals, or feedback.

5. User Stories and Acceptance Criteria

User Type	Functional Requirement (Epic)	User Story / Task	Acceptance Criteria	Priority	Release
Employee	Authentication	As an employee, I can register and log in securely.	Valid credentials allow access; invalid credentials are rejected; public registration cannot assign HR or Manager roles.	High	Sprint 1
Employee	Profile Management	As an employee, I can view and update my permitted profile details.	Valid changes are saved and appear when the profile is reopened.	High	Sprint 1
Employee	Attendance	As an employee, I can record attendance and view my history.	Attendance actions are saved with server-side timestamps and the history shows correct records.	High	Sprint 1
Employee	Leave Application	As an employee, I can submit a leave request and track its status.	The request is stored and its status changes after an authorized review.	High	Sprint 1
Manager	Team Management	As a manager, I can view only employees assigned to my team.	Team results are limited to the manager's authorized team.	High	Sprint 2
Manager	Leave Review	As a manager, I can approve or reject eligible team leave requests.	The decision is stored, status is updated, and the employee can see the result.	High	Sprint 2
Manager	Team Attendance	As a manager, I can review attendance summaries for my team.	Only authorized team attendance records are returned.	Medium	Sprint 2
HR	Employee Management	As HR, I can create, view, update, and manage employee records.	Authorized HR actions persist correctly and unauthorized roles are denied.	High	Sprint 1
HR	Manager Management	As HR, I can create and manage manager accounts.	The manager record and corresponding user identity are associated correctly, and credentials are securely hashed.	High	Sprint 2

HR	Attendance Administration	As HR, I can review attendance records across the organization.	HR can access organization-wide records and filters return expected results.	High	Sprint 2
HR	Announcements	As HR, I can publish announcements for users.	Published announcements are saved and visible to the intended audience.	Medium	Sprint 2
HR	Dashboard & Analytics	As HR, I can view summary metrics for employees, attendance, and leave.	Dashboard totals match stored records and respect access permissions.	Medium	Sprint 2
System	Notifications	As a user, I can receive notifications about relevant actions.	Relevant events create notifications for the intended recipient and can be viewed.	Medium	Sprint 2
Authorized User	Performance Management	As an authorized user, I can view or manage performance records permitted by my role.	Only permitted records and actions are available to each role.	Medium	Sprint 2

6. Security and Validation Notes

- Passwords must be stored as hashes; plaintext passwords must never be persisted.
- Public registration should create Employee accounts only. HR and Manager privileges must be granted through a protected administrative flow.
- Protected APIs must verify JWTs and enforce role-based authorization on the server, not only in the React interface.
- Managers should only access team members and requests they are authorized to manage.
- Validate incoming data, handle errors consistently, and keep secrets in environment variables rather than source control.
- Do not run destructive seed scripts against a database containing project data.

7. Scope Note

This document describes StaffSync's intended system flows and user stories based on the project scope. It does not claim that every feature has been tested or is currently deployed. Verify implemented routes, role permissions, and acceptance criteria against the current repository and running application before submission.